

Question Number	Answer	Additional guidance	Mark
5(a)	A description that includes two of the following points: • analysis of an aspect of phenotype (1) • analysis of (the sequences in) biological molecules (1) • (to see if) there are (many) differences (to existing types of bacteria) (1)	e.g. {named / described} aspect of physical appearance / analysis of growth characteristics / biochemical tests accept compare named aspect of structure / named organelle / ribosomes accept see if other bacteria survive in those conditions ignore behaviour e.g. looking at (circular) DNA, genetic material, proteins accept molecular phylogeny / genetic profiling / create genetic fingerprint ignore compare to other bacteria species unqualified ignore similarities	(2)

Question Number	Answer	
*5(b)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • description of growth rate for {one / both} bacteria e.g. growth rate increased and then decreased • growth rate of bacteria B was higher than growth rate of bacteria A (at optimum temperatures) • Bacteria A {can survive in temperatures between -4-21°C / is not tolerant of temperatures above 21°C} whereas bacteria B {can survive in temperatures between 66-105°C / is not tolerant of temperatures below 66°C} • Bacteria A {optimum / highest growth} rate is around 10°C whereas bacteria B {optimum / highest growth} rate is around 95°C • explanation of the increase and decrease in growth rate as temperature increases e.g. linkage to enzyme activity / optimum temperature (linked to highest growth) / denaturation / death of bacteria • explanation linking enzyme activity linked to how it would affect growth rate of bacteria e.g. new cell wall production / production of cell structures / protein synthesis / metabolic reactions / respiration linked to energy available for growth • explanation of how bacteria B can survive in the hot temperatures, e.g. {proteins / enzymes} that are thermostable • The percentage survival of bacteria reduces (when water was removed) / negative correlation • Bacteria C survived for longer (without water) / more bacteria C survived (than mutated bacteria) • The percentage survival of bacteria C reduced by over 70% after one day and no bacteria survived after 9 days / the percentage survival of mutated bacteria C reduced by 10% after one day and 60% bacteria C survived after 15 days • error bars overlap so no significant difference • Mutated bacteria C had a mutation that did not protect it from dehydration e.g. {thinner / no} capsule / thinner cell wall / can't do reactions which produce water • mutation could lead to different protein produced / mutation could lead to change in cell structure / description of how mutations change protein structure • explanation of why bacteria need water e.g. {hydrolysis / respiration / chemical reactions / metabolic reactions / solvent / osmosis}	(6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	description of data from graphs only 1 mark = description of one graph 2 marks = description of both graphs
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	all level 1 plus 3 marks = basic explanation for one graph 4 marks = detailed explanation for one graph OR basic explanation for both graphs
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	all level 2 plus detailed explanation for both graphs 5 marks = detailed explanation for one graph AND basic explanation for one graph 6 marks = detailed explanation for both graphs